Selective Context: Prompt Compression for Efficient LLM-Based News Understanding

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PEGAI Course Project Report | November 2026

Abstract –

Large language models (LLMs) can perform news summarisation, question answering, and response generation using long textual contexts. However, processing unnecessary context increases input-token usage and may introduce irrelevant information. This project studies phrase-level context compression using a language-model-based approach and evaluates its effects across four downstream tasks: news summarisation, question answering (QA), response generation, and original-context reconstruction. Phrase importance is estimated using a local language-model-based self-information signal, and low-importance phrases are removed at configured compression levels of 20%, 35%, 50%, 65%, and 80%, alongside an uncompressed baseline. The summarisation experiment uses 29 articles and reports ROUGE and BERTScore; the QA and response-generation experiments use 10-article subsets; reconstruction is evaluated on the same 10 articles. Results show that light compression (20%) can preserve or slightly improve summarisation scores while reducing input tokens by approximately 18.18% on average. More aggressive compression consistently reduces QA performance, while response-generation BERTScore remains comparatively stable across the tested conditions. Reconstruction metrics are mixed and do not support a clear conclusion. The findings indicate that the usefulness of selective context compression is task-dependent and that moderate compression can offer a practical efficiency–quality trade-off for summarisation, while aggressive compression risks removing information required by other tasks.

Index Terms –

Context compression, selective context, large language models, news summarisation, question answering, response generation, information preservation.

I. INTRODUCTION

Large language models are increasingly used to transform unstructured news articles into concise summaries, answer questions about article content, and produce context-grounded responses. These applications often pass substantial portions of source documents to a model, even when only a subset of the text is relevant to the requested task. Longer prompts increase token consumption and may increase inference cost and latency. They can also expose the model to distracting or redundant material.

Context compression attempts to reduce this burden by removing less useful parts of an input while retaining information that supports downstream reasoning. This project investigates phrase-level context compression using language-model self-information to estimate phrase importance and evaluates its effects in a BBC News summarisation and understanding workflow across multiple tasks.

The central question is: How does phrase-level context compression affect the quality and efficiency of LLM-based news understanding? Rather than treating compression as a single summarisation-only operation, the project evaluates four complementary tasks to explore whether information retained for one task remains useful for others.

A. Objectives

The project has four objectives: (1) implement phrase-level selective context compression with multiple configured removal ratios; (2) measure summarisation quality and input-token reduction; (3) evaluate the effect of compression on QA and response generation; and (4) assess whether a model can reconstruct an original article from its compressed representation.

B. Contributions

The project provides an experimental implementation and comparison across four downstream tasks, a common set of compression conditions, metric-based evaluation, and paired statistical comparisons where available. The work focuses on the design, implementation, and evaluation of the compression workflow for the selected news-understanding tasks.

II. RELATED WORK

Context compression methods aim to reduce the length of model inputs while retaining information relevant to a downstream task. Phrase-level approaches estimate the relative informativeness of text segments and selectively remove lower-priority content. This project examines such a strategy in a news-understanding setting and compares the impact of phrase removal across multiple downstream tasks.

Automatic summarisation is commonly evaluated using lexical-overlap measures such as ROUGE [1], while semantic similarity measures such as BERTScore [2] compare contextual token representations. For extractive or extractive-style QA, Exact Match (EM) and token-level F1 are widely used to assess answer overlap. Since each metric captures a different aspect of output quality, this study reports multiple measures and avoids treating any single score as a complete measure of correctness.

III. DATASET AND EXPERIMENTAL DESIGN

A. News Dataset

The source corpus is the RealTimeData BBC News March 2023 dataset, accessed through its Hugging Face dataset release. The local project dataset contains 1,892 article records. The summarisation experiment used 29 articles. QA, response generation, and context reconstruction used a smaller, aligned subset of 10 articles to support comparison across these tasks. The QA dataset contains 30 question–answer pairs (three per article), and the response-generation set contains 20 dialogue examples.

B. Compression Conditions

Six conditions were evaluated: an uncompressed baseline (0%) and configured phrase-removal ratios of 20%, 35%, 50%, 65%, and 80%. The ratio denotes the configured proportion of phrases targeted for removal; it does not denote the percentage of context retained. The realised token reduction can differ from the configured phrase-removal ratio because phrases have different token lengths.

C. Compression Method

Phrase-level importance is estimated using self-information from a locally executed GPT-2 language model. The project applies the existing NewsContextManager implementation with phrase-level masking. For each input article, the compressor estimates phrase informativeness and removes lower-importance phrases according to the selected condition. The resulting compressed context is then supplied to the task-specific model. The baseline uses the original article context without this compression step.

D. Models and Task Configuration

TABLE I
MODELS AND TASK CONFIGURATION

Task	Model / setup	Evaluation sample
Summarisation	sshleifer/distilbart-cnn-6-6; GPT-2 phrase scoring	29 articles × 6 conditions = 174 outputs
Question answering	valhalla/t5-base-qa-qg-hl for question generation; deepset/roberta-base-squad2 for QA	30 Q/A pairs × 6 conditions = 180 predictions
Response generation	Task-specific generation pipeline; semantic evaluation with BERTScore	20 dialogue examples from 10 articles
Context reconstruction	google/flan-t5-base; maximum input/output length configured at 512 tokens	10 articles × 6 conditions = 60 reconstructions

Experiments were run locally on CPU. The summarisation and reconstruction evaluation used BERTScore with distilroberta-base. ROUGE variants were used for summarisation and reconstruction; QA used Exact Match and token F1. Exact model decoding settings and hardware throughput were not treated as controlled benchmark variables in this report.

IV. METHODOLOGY

A. News Summarisation

For each of the 29 selected articles, the original article and five compressed variants were passed to the DistilBART summarisation model. Generated summaries were compared with the dataset reference summaries using ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore F1. Mean input-token reduction was also computed relative to the uncompressed article. The total experiment comprised 174 generated summaries.

B. Question Answering

A set of 30 question–answer pairs was prepared from 10 articles. Each question was evaluated against the original context and each compressed context using the RoBERTa SQuAD2 QA model. Exact Match and token-level F1 were calculated against the supplied reference answers. The QA model input was limited to 512 tokens; consequently, long baseline articles could be truncated. This limit is an important constraint when interpreting the baseline and compressed-context comparison.

C. Response Generation

The response-generation experiment evaluated 20 dialogue examples associated with 10 source articles. For each compression condition, the task pipeline generated a response from the available dialogue/article context. BERTScore F1 was used to compare generated responses with the reference responses. The experiment is small and the response references are concise; therefore, semantic similarity scores are interpreted descriptively.

D. Original-Context Reconstruction

The reconstruction task supplied compressed article contexts to FLAN-T5-base and asked the model to reconstruct the original article. Ten articles were evaluated under all six conditions, producing 60 reconstructions. Evaluation included BERTScore, ROUGE, BLEU, and METEOR. Both model input and generated output were capped at 512 tokens. Since full news articles can exceed these limits, this task measures constrained reconstruction rather than unrestricted recovery of the complete source article.

E. Statistical Analysis

Paired Wilcoxon signed-rank tests were used for comparisons between compressed conditions and the uncompressed baseline where paired observations were available. Holm correction was applied to control the family-wise error rate across multiple comparisons. Statistical significance is reported only where the adjusted p-value crossed the 0.05 threshold. Small sample sizes, especially for the 10-article tasks, limit statistical power.

V. RESULTS

A. Summarisation Results

Table II reports mean summarisation scores. At 20% configured phrase removal, ROUGE-1 increased from 0.2664

to 0.2762, ROUGE-2 from 0.0973 to 0.1116, ROUGE-L from 0.1940 to 0.2139, and BERTScore F1 from 0.8955 to 0.8971. This condition also reduced input tokens by approximately 18.18% on average. At higher compression levels, scores generally declined, with the largest deterioration at 80%.

TABLE II
SUMMARISATION RESULTS

Removal	R-1	R-2	R-L	BERT F1	Token ↓
0%	0.2664	0.0973	0.1940	0.8955	0.00%
20%	0.2762	0.1116	0.2139	0.8971	18.18%
35%	0.2481	0.0837	0.1850	0.8913	29.81%
50%	0.2269	0.0631	0.1699	0.8842	44.11%
65%	0.2082	0.0488	0.1460	0.8791	61.36%
80%	0.1442	0.0222	0.1110	0.8668	79.20%

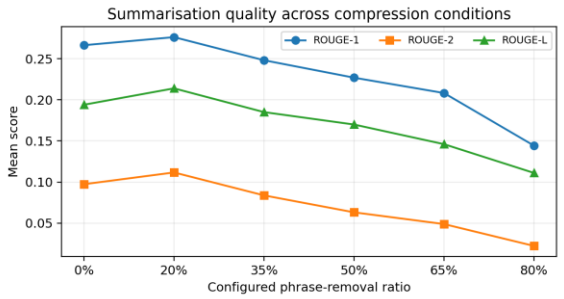


Fig. 1. Mean ROUGE scores across summarisation compression conditions.

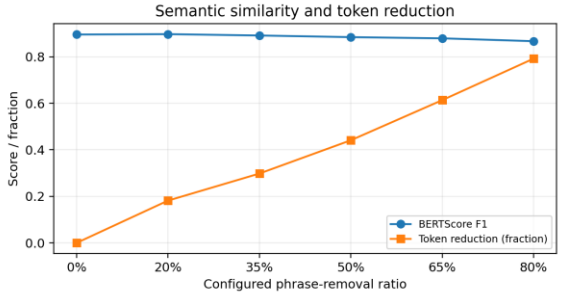


Fig. 2. BERTScore F1 and measured token reduction across conditions.

Paired statistical comparisons indicated no significant quality difference at 20% and 35% after correction. At 50%, the adjusted BERTScore comparison showed a significant decline while ROUGE differences did not remain significant after correction. At 65% and 80%, declines were significant across the reported quality metrics. Thus, the strongest evidence for a useful trade-off is at light compression, not at aggressive compression.

B. Question-Answering Results

TABLE III
QUESTION-ANSWERING RESULTS

Removal	Exact Match	Token F1
0%	63.33%	87.16%
20%	56.67%	83.56%
35%	40.00%	68.06%
50%	36.67%	57.49%
65%	36.67%	53.88%
80%	20.00%	32.38%

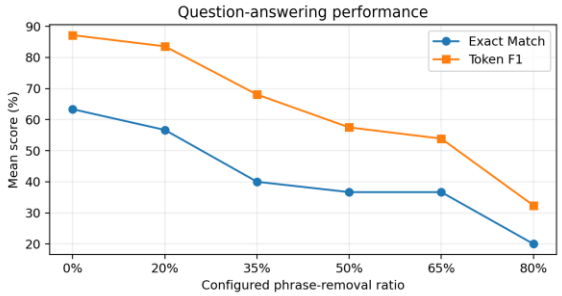


Fig. 3. QA Exact Match and token F1 across compression conditions.

QA performance decreased as compression increased. The 20% and 35% conditions did not show statistically significant differences from baseline after Holm correction. Token F1 declines at 50% and 65% were significant after correction, while Exact Match did not cross the adjusted threshold at those levels. At 80%, both Exact Match and token F1 declined significantly. These results suggest that QA depends on details that may be removed even when the compressed context remains adequate for generating a broad summary.

The QA experiment is exploratory: it contains 30 question–answer pairs from 10 articles, and the QA model's 512-token maximum input length constrained long contexts. Reference answers were manually reviewed, but they should not be treated as independently annotated ground truth.

C. Response-Generation Results

TABLE IV
RESPONSE-GENERATION RESULTS

Removal	BERTScore F1
0%	0.8599
20%	0.8586
35%	0.8644
50%	0.8565
65%	0.8548
80%	0.8537

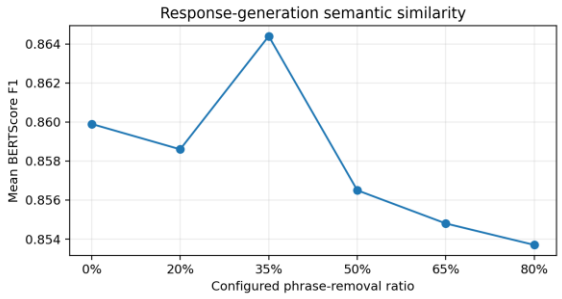


Fig. 4. Response-generation BERTScore F1 across compression conditions.

Response-generation BERTScore remained within a relatively narrow range (0.8537–0.8644). The highest mean score occurred at 35% removal (0.8644), while the baseline was 0.8599. Paired comparisons with Holm correction did not identify statistically significant differences. The measured reduction in history characters ranged from approximately 52.3% at the configured 20% condition to 70.8% at 80%; configured phrase removal and character reduction are different quantities. The small sample and lack of significant

differences mean that stability should not be interpreted as proof of equivalence.

D. Original-Context Reconstruction Results

TABLE V
ORIGINAL-CONTEXT RECONSTRUCTION RESULTS

Removal	BERT F1	ROUGE-1	ROUGE-L
0%	0.8600	0.0595	0.0520
20%	0.8417	0.0451	0.0410
35%	0.8428	0.0494	0.0464
50%	0.8411	0.0595	0.0477
65%	0.8503	0.0949	0.0754
80%	0.7545	0.1409	0.1198

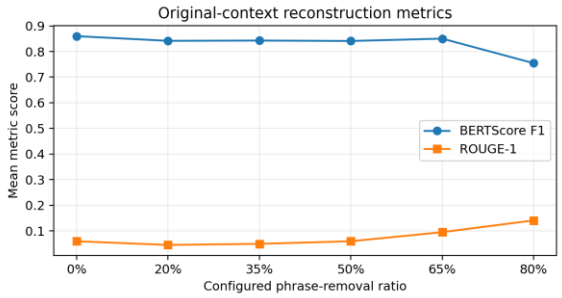


Fig. 5. Reconstruction metrics; lexical and semantic measures show different trends.

The reconstruction metrics disagree. BERTScore F1 remains near 0.84 through 65% removal before falling to 0.7545 at 80%, whereas ROUGE-1 and ROUGE-L rise at higher compression. This pattern can occur when a generated reconstruction shares common wording or structure with a reference while failing to preserve its full semantic content, or when metric behaviour is affected by output length. No metric-by-condition comparison remained significant after Holm correction. Given only 10 articles and the 512-token input/output caps, the experiment is inconclusive and should not be used to claim that original articles can be reliably recovered from heavily compressed contexts.

VI. DISCUSSION

Across the four tasks, compression quality is task-dependent. Summarisation tolerated light phrase removal and showed a small improvement at 20%, potentially because removing redundant or low-information phrases can leave a more focused input. The benefit did not persist at higher ratios: the summarisation scores declined as more phrases were removed. QA was more sensitive to aggressive compression, consistent with its dependence on specific entities, dates, quantities, and relations. Response generation showed comparatively stable BERTScore, but the sample was too small to establish equivalence or generalise to other dialogue settings. Reconstruction was the least conclusive task because the model had to regenerate long source text under a strict token budget, and the evaluation metrics moved in different directions.

The results therefore do not support a universal optimal compression ratio. For the tested summarisation setup, 20% is the most promising tested condition because it reduced input tokens by about 18% while maintaining or slightly improving average quality. For QA, a conservative compression policy is preferable when exact factual details matter. A production system should ideally select a compression level based on the downstream task and preserve named entities, numbers, dates, and other task-critical spans.

VII. THREATS TO VALIDITY AND LIMITATIONS

Dataset coverage is limited: 29 articles were used for summarisation and 10 for each of the other tasks. These samples do not establish performance across news domains, writing styles, languages, or longer documents. The QA questions and response examples are limited in number, and their references are not a large independently annotated benchmark.

Model and context limits affect comparability. The QA model accepts at most 512 input tokens, and reconstruction used 512-token input and output limits. These constraints can truncate the baseline and compressed contexts and prevent complete article reconstruction. The experiments were run on CPU, and measured generation times were variable; therefore, no robust latency improvement claim is made. Token reduction is a direct efficiency proxy, not a complete measurement of wall-clock speed, monetary cost, or energy consumption.

Automatic metrics are imperfect proxies for factual correctness and usefulness. ROUGE rewards lexical overlap, BERTScore estimates semantic similarity, and QA EM/F1 measure answer overlap. Metric disagreement in reconstruction illustrates why human evaluation and factuality checks are necessary. Finally, the project evaluates a phrase-level context compression workflow for the selected tasks; it does not claim a new state-of-the-art result.

VIII. CONCLUSION AND FUTURE WORK

This project evaluated phrase-level selective context compression for four news-understanding tasks. Light compression produced the clearest positive trade-off in summarisation: the 20% condition reduced input tokens by approximately 18.18% while maintaining or slightly improving mean ROUGE and BERTScore. QA quality declined progressively with stronger compression, and response-generation scores were comparatively stable but statistically inconclusive. Original-context reconstruction produced conflicting metric trends and remains inconclusive under the current sample and token limits.

Future work should expand the article sample, use independently annotated QA and response references, evaluate longer-context models, include human assessments of factual consistency and completeness, and measure latency, token cost, and energy under controlled conditions. A task-aware compressor that protects named entities, numerical facts, dates, and relations is a promising extension.

REFERENCES

[1] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in Text Summarization Branches Out, 2004, pp. 74–81.

[2] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in Proc. ICLR, 2020.